Table-top lab · guide

Spider Webs

Look-don't-touch: watch webs from a step back. Outdoor dew walks are best on cool mornings. If anyone is afraid of spiders, keep the lesson on pictures and sketches. Hedge aloud with your child: many orb webs work this way, but spiders build other shapes too.

Try 1 · Which lines are sticky on an orb web?

From a few steps away, sketch a web you can see. Mark the hub, three spokes, and one loop of the spiral. Do not touch the silk.

Look for: On many orb webs, the spokes (radii) and the small hub are dry and strong. The outer spiral often holds tiny glue beads. Spiders walk the dry lines and feel prey on the sticky spiral. Not every web is an orb, and not every spiral is sticky the same way.

Try 2 · How can silk catch without glue?

On your sketch, label one spiral loop 'maybe glue' and write one question you still have about silk.

Look for: Some spiders make capture silk with glue droplets (ecribellate silk). Others comb out a dry, woolly silk with a special plate called a cribellum (cribellate silk). The woolly silk snags legs and wings without sticky glue. Scientists still study exactly how each kind works best.

Try 3 · How does a spider feel a tiny tug?

Tap a table lightly with a pencil, then harder. Tell which tap felt bigger through your hand. That is only a toy model of vibration, not a real web.

Look for: Silk carries vibrations. Many spiders keep legs on the radii so a buzz or tug can travel to them. Tiny sensors in the legs help them tell a breeze from a struggling insect. They usually do not need sticky feet on every line because they know the dry paths.

Try 4 · Why does dew make a web easy to see?

On a cool morning, look for a web on a fence or bush. Count how many spiral loops you can see before the dew fades. Stay back and do not touch.

Look for: Cool mornings leave tiny water drops on silk. The drops catch light, so the web that was nearly invisible overnight suddenly shows its pattern. As the air warms, many drops shrink or fall. Dew helps us look; it is not glue from the spider.

Print the lab sheet and this guide separately.